

1. Which data type is used to store whole numbers in C?

- a) float
- b) int
- c) char
- d) double

Answer: b

2. What does a compiler do?

- a) Runs the program
- b) Converts source code into machine code
- c) Stores data in memory
- d) Designs the hardware

Answer: b

3. Which symbol terminates a statement in C?

- a) :
- b) .
- c) ;
- d) ,

Answer: c

4. What is the value of $7 \% 3$ in C?

- a) 1
- b) 2
- c) 3
- d) 4

Answer: a

5. Which loop is guaranteed to execute at least once?

- a) for
- b) while
- c) do-while
- d) if

Answer: c

6. What is the typical size of an int in a 32-bit C compiler?

- a) 1 byte
- b) 2 bytes
- c) 4 bytes
- d) 8 bytes

Answer: c

7. Which function reads a line of text from standard input?

- a) scanf
- b) gets
- c) printf
- d) getline

Answer: b

8. What is an array in C?

- a) A single variable
- b) A collection of variables of the same type
- c) A function that returns multiple values
- d) A pointer to a structure

Answer: b

9. Which operator accesses a member of a structure using its name?

- a) .
- b) ->
- c) &
- d) *

Answer: a

10. What does the ++ operator do to a variable?

- a) Decrements it by 1
- b) Doubles its value
- c) Increments it by 1
- d) Compares it to zero

Answer: c

11. Explain the difference between a compiler and an interpreter.

(Write your answer in the space provided.)

12. Write a short note on the advantages of using functions in programming.

(Write your answer in the space provided.)